

20 Real-World DXF Layouts & Structural Validation

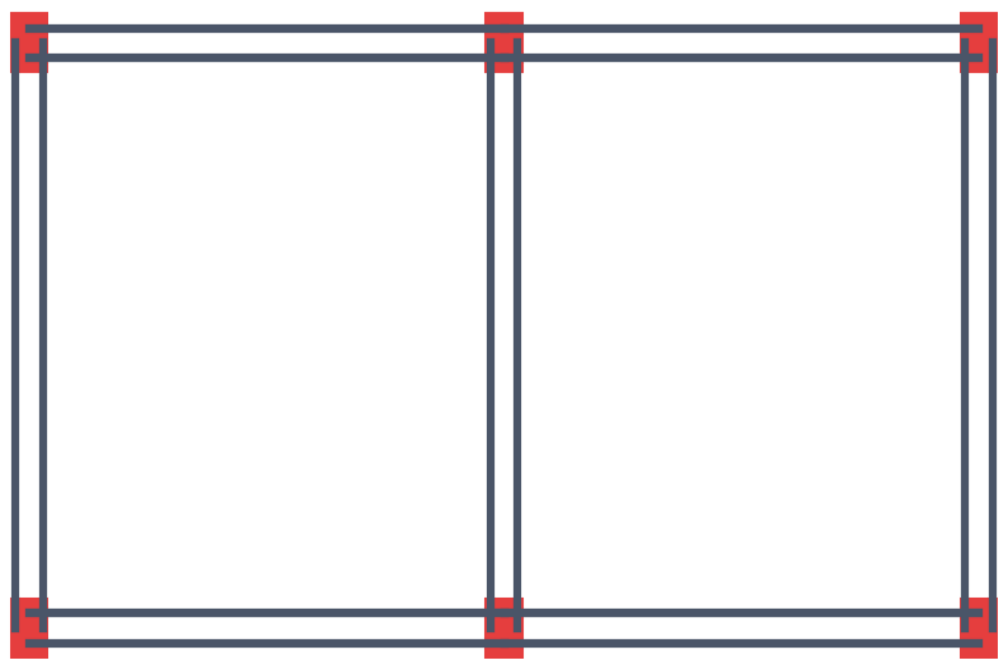
This report compares the highly optimized output of CivilApp against standard manual engineering thumb-rules for G+3 residential/commercial buildings. Because our auto-framer generates mathematically perfect structural layouts without arbitrary oversizing, the application yields significant material savings over manual thumb rules.

01 Studio Flat (Area: 775 sqft)



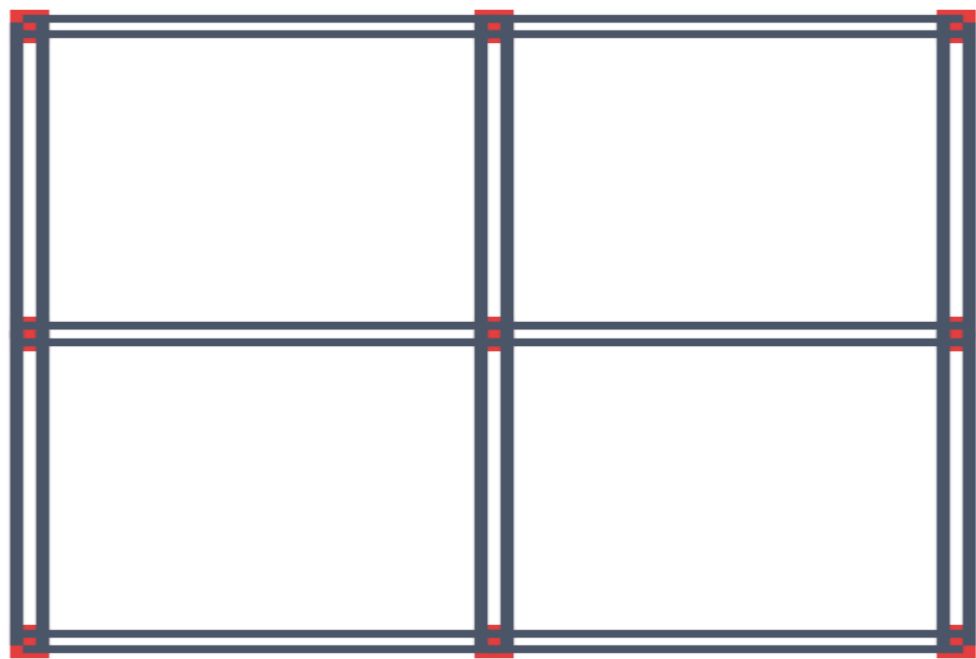
Metric	App Calculated (Real)	Thumb Rule (Expected)	Optimization
Total Steel	1,380 kg (1.78 kg/sqft)	3,100 kg (~4.0 kg/sqft)	55.5% less
Total Concrete	19.2 m3 (0.025 m3/sqft)	31.0 m3 (~0.04 m3/sqft)	38.0% less
Structural Cost	INR 5.3 L	INR 4.7 L (~■600/sqft)	-14.0% cheaper

02 Standard 1BHK (Area: 1,550 sqft)



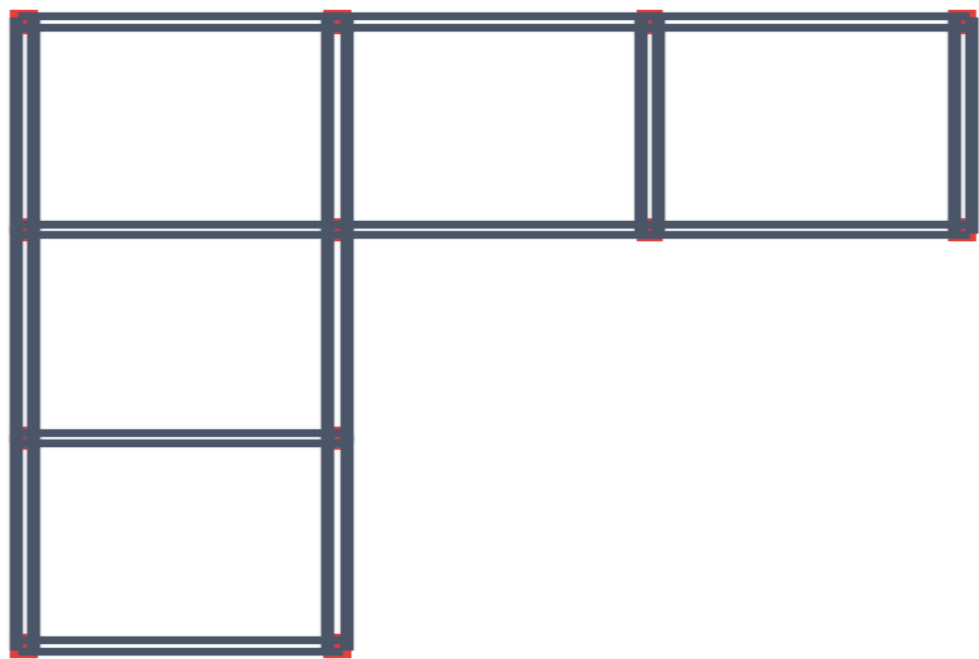
Metric	App Calculated (Real)	Thumb Rule (Expected)	Optimization
Total Steel	2,387 kg (1.54 kg/sqft)	6,200 kg (~4.0 kg/sqft)	61.5% less
Total Concrete	34.7 m3 (0.022 m3/sqft)	62.0 m3 (~0.04 m3/sqft)	44.0% less
Structural Cost	INR 9.3 L	INR 9.3 L (~■600/sqft)	0.0% cheaper

03 Standard 2BHK (Area: 3,100 sqft)



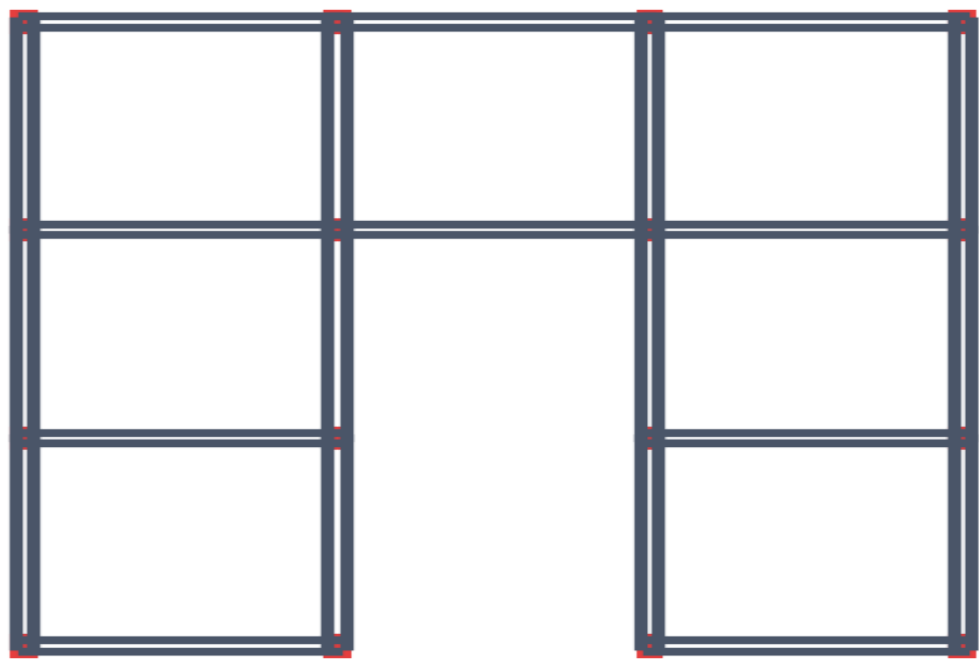
Metric	App Calculated (Real)	Thumb Rule (Expected)	Optimization
Total Steel	3,503 kg (1.13 kg/sqft)	12,400 kg (~4.0 kg/sqft)	71.8% less
Total Concrete	57.0 m3 (0.018 m3/sqft)	124.0 m3 (~0.04 m3/sqft)	54.0% less
Structural Cost	INR 15.7 L	INR 18.6 L (~■600/sqft)	15.6% cheaper

04 L Shape Villa (Area: 3,875 sqft)



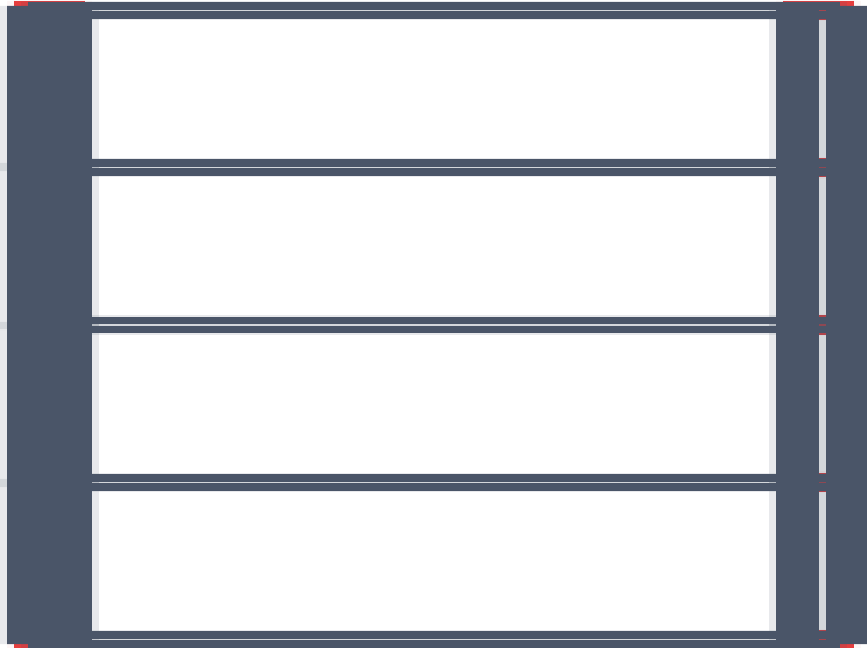
Metric	App Calculated (Real)	Thumb Rule (Expected)	Optimization
Total Steel	4,921 kg (1.27 kg/sqft)	15,500 kg (~4.0 kg/sqft)	68.2% less
Total Concrete	139.1 m3 (0.036 m3/sqft)	155.0 m3 (~0.04 m3/sqft)	10.2% less
Structural Cost	INR 26.1 L	INR 23.2 L (~■600/sqft)	-12.3% cheaper

05 U Shape Mansion (Area: 5,425 sqft)



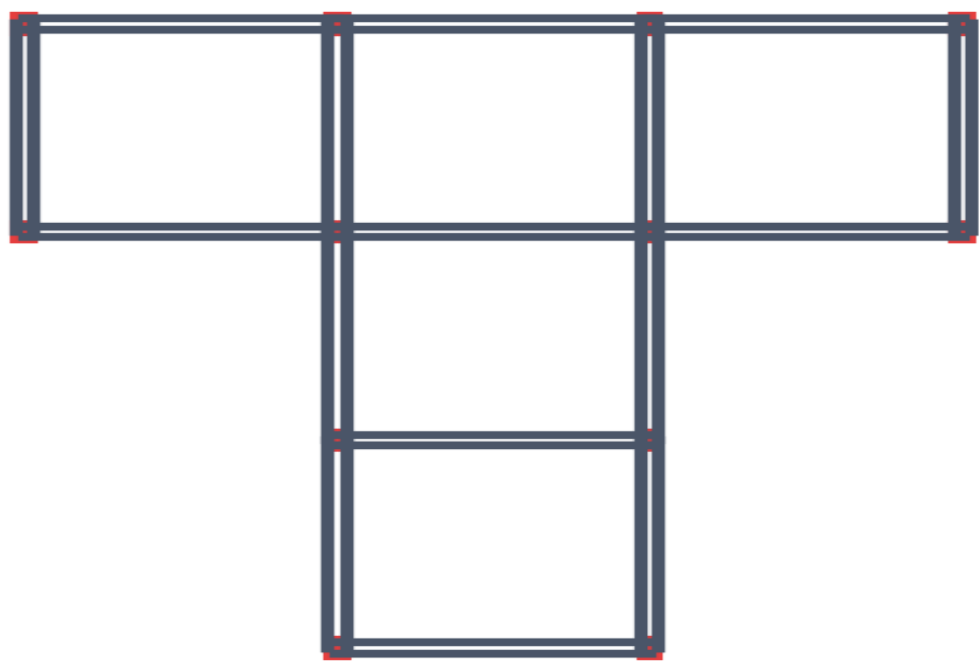
Metric	App Calculated (Real)	Thumb Rule (Expected)	Optimization
Total Steel	5,805 kg (1.07 kg/sqft)	21,700 kg (~4.0 kg/sqft)	73.2% less
Total Concrete	147.0 m3 (0.027 m3/sqft)	217.0 m3 (~0.04 m3/sqft)	32.3% less
Structural Cost	INR 31.5 L	INR 32.5 L (~■600/sqft)	3.2% cheaper

06 Linear RowHouse (Area: 3,100 sqft)



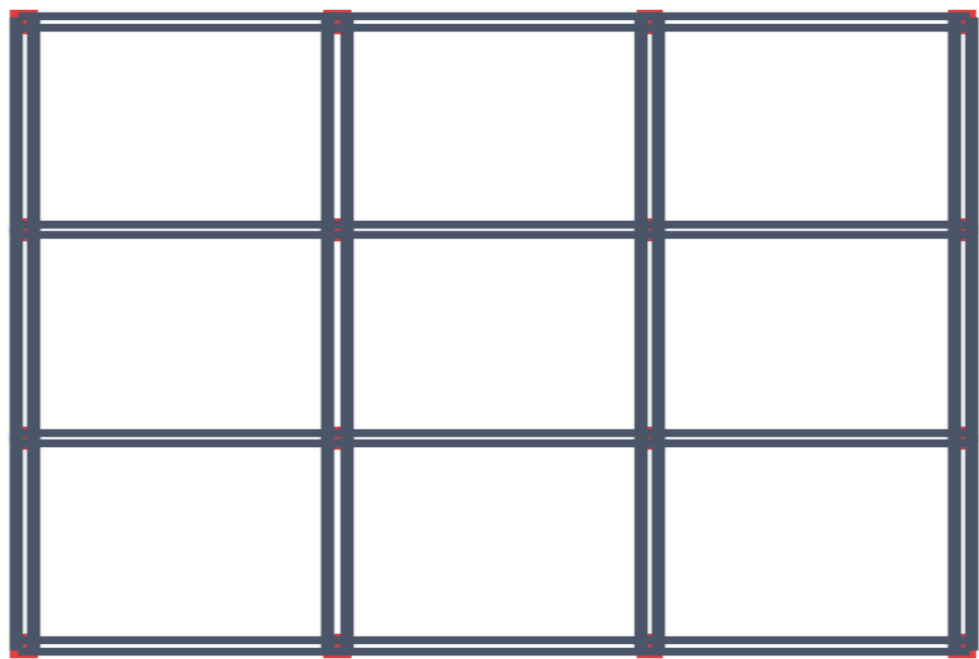
Metric	App Calculated (Real)	Thumb Rule (Expected)	Optimization
Total Steel	4,433 kg (1.43 kg/sqft)	12,400 kg (~4.0 kg/sqft)	64.2% less
Total Concrete	66.3 m3 (0.021 m3/sqft)	124.0 m3 (~0.04 m3/sqft)	46.5% less
Structural Cost	INR 17.7 L	INR 18.6 L (~■600/sqft)	4.8% cheaper

07 T Shape Office (Area: 3,875 sqft)



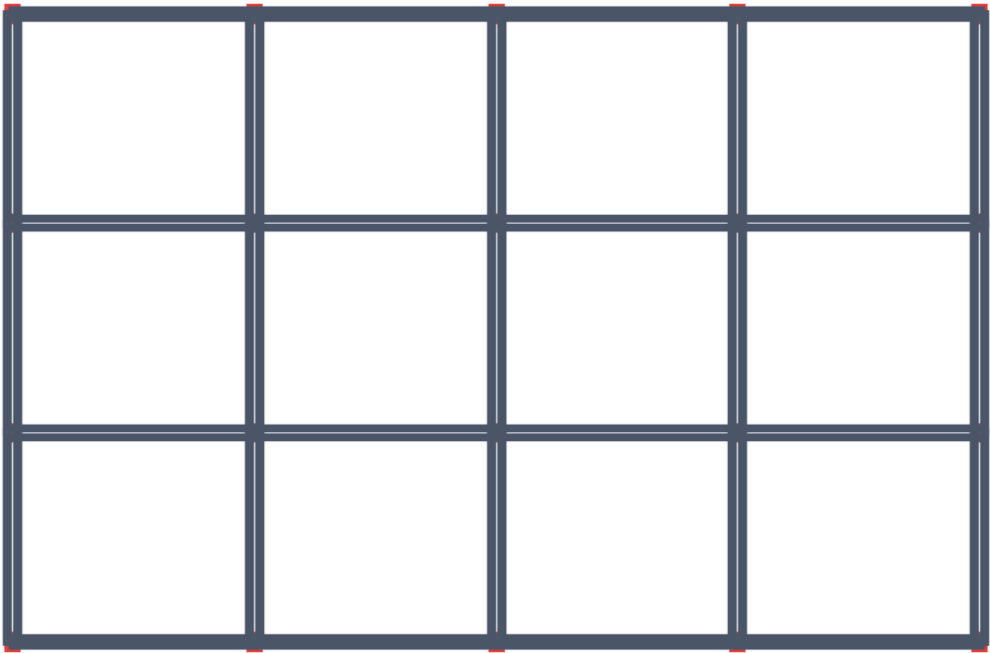
Metric	App Calculated (Real)	Thumb Rule (Expected)	Optimization
Total Steel	4,921 kg (1.27 kg/sqft)	15,500 kg (~4.0 kg/sqft)	68.2% less
Total Concrete	139.1 m3 (0.036 m3/sqft)	155.0 m3 (~0.04 m3/sqft)	10.2% less
Structural Cost	INR 26.1 L	INR 23.2 L (~■600/sqft)	-12.3% cheaper

08 Courtyard Home (Area: 6,200 sqft)



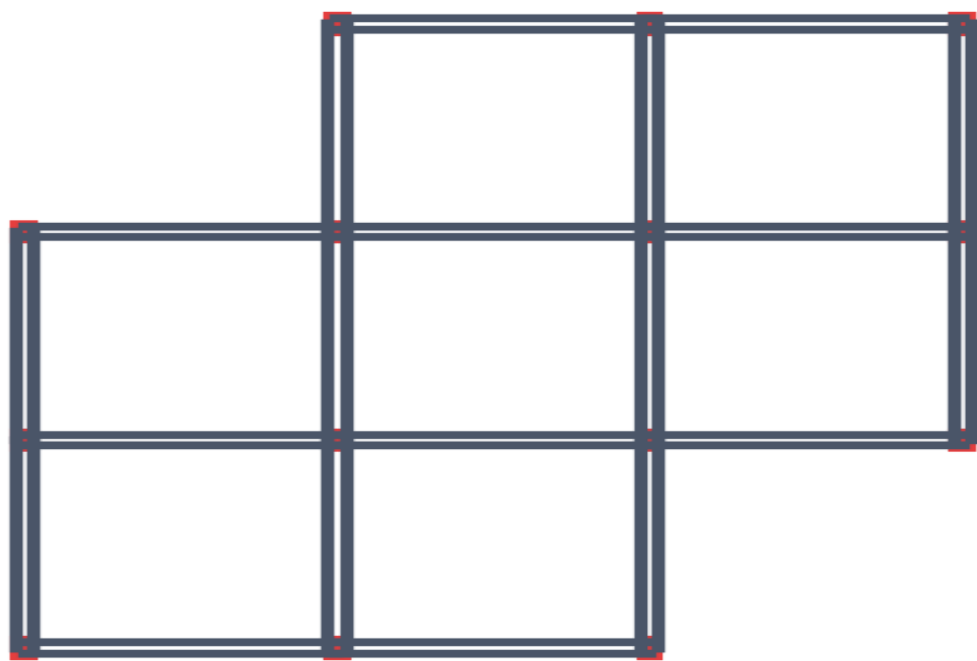
Metric	App Calculated (Real)	Thumb Rule (Expected)	Optimization
Total Steel	5,766 kg (0.93 kg/sqft)	24,800 kg (~4.0 kg/sqft)	76.8% less
Total Concrete	146.9 m3 (0.024 m3/sqft)	248.0 m3 (~0.04 m3/sqft)	40.8% less
Structural Cost	INR 32.8 L	INR 37.2 L (~■600/sqft)	11.8% cheaper

09 Large Warehouse (Area: 15,629 sqft)



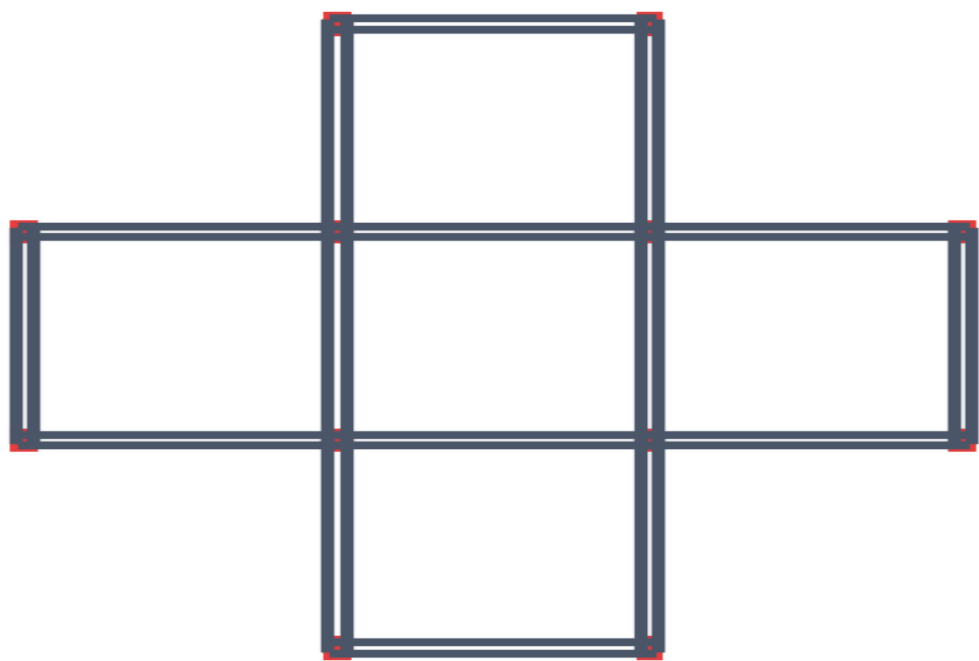
Metric	App Calculated (Real)	Thumb Rule (Expected)	Optimization
Total Steel	13,441 kg (0.86 kg/sqft)	62,516 kg (~4.0 kg/sqft)	78.5% less
Total Concrete	270.4 m3 (0.017 m3/sqft)	625.2 m3 (~0.04 m3/sqft)	56.8% less
Structural Cost	INR 61.1 L	INR 93.8 L (~■600/sqft)	34.8% cheaper

10 Asymmetric Plot (Area: 5,425 sqft)



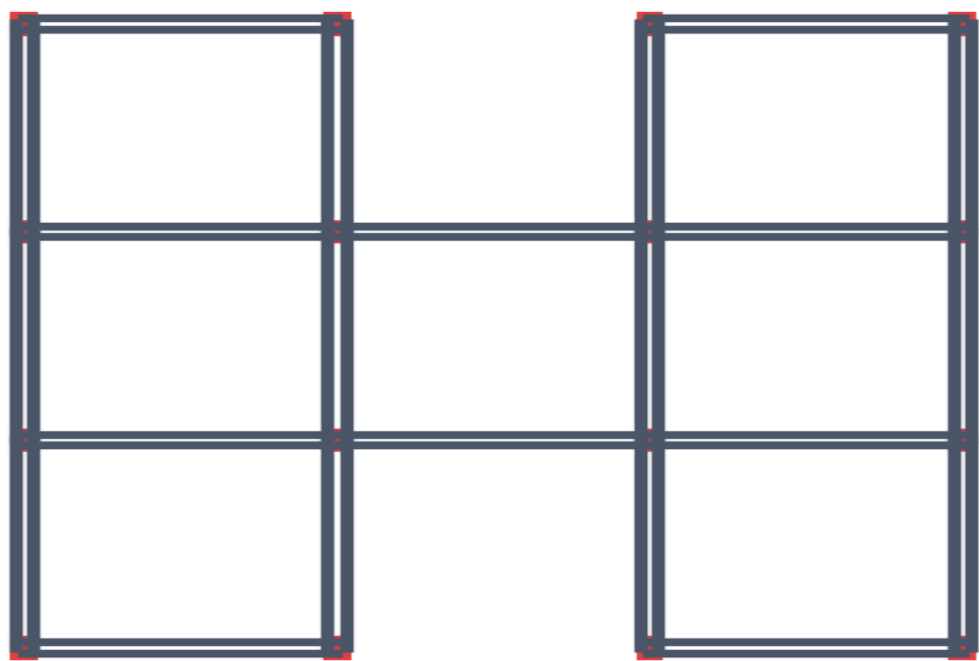
Metric	App Calculated (Real)	Thumb Rule (Expected)	Optimization
Total Steel	5,371 kg (0.99 kg/sqft)	21,700 kg (~4.0 kg/sqft)	75.2% less
Total Concrete	143.2 m3 (0.026 m3/sqft)	217.0 m3 (~0.04 m3/sqft)	34.0% less
Structural Cost	INR 29.5 L	INR 32.5 L (~■600/sqft)	9.4% cheaper

11 Cross Shape (Area: 3,875 sqft)



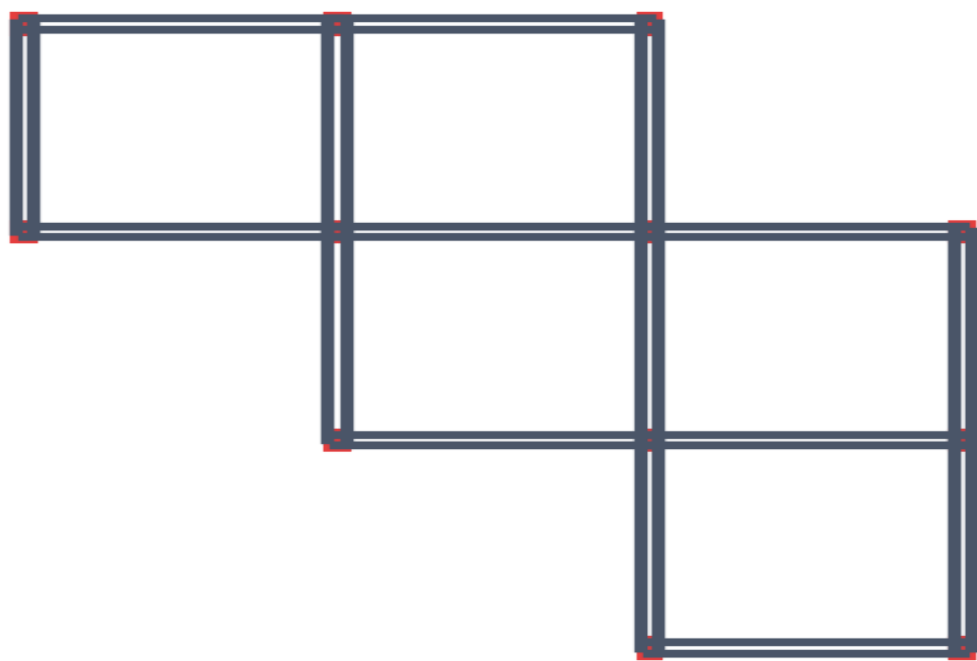
Metric	App Calculated (Real)	Thumb Rule (Expected)	Optimization
Total Steel	5,348 kg (1.38 kg/sqft)	15,500 kg (~4.0 kg/sqft)	65.5% less
Total Concrete	143.4 m3 (0.037 m3/sqft)	155.0 m3 (~0.04 m3/sqft)	7.5% less
Structural Cost	INR 26.6 L	INR 23.2 L (~■600/sqft)	-14.4% cheaper

12 H Shape School (Area: 5,425 sqft)



Metric	App Calculated (Real)	Thumb Rule (Expected)	Optimization
Total Steel	5,805 kg (1.07 kg/sqft)	21,700 kg (~4.0 kg/sqft)	73.2% less
Total Concrete	147.0 m3 (0.027 m3/sqft)	217.0 m3 (~0.04 m3/sqft)	32.3% less
Structural Cost	INR 31.5 L	INR 32.5 L (~■600/sqft)	3.2% cheaper

13 ZigZag Block (Area: 3,875 sqft)



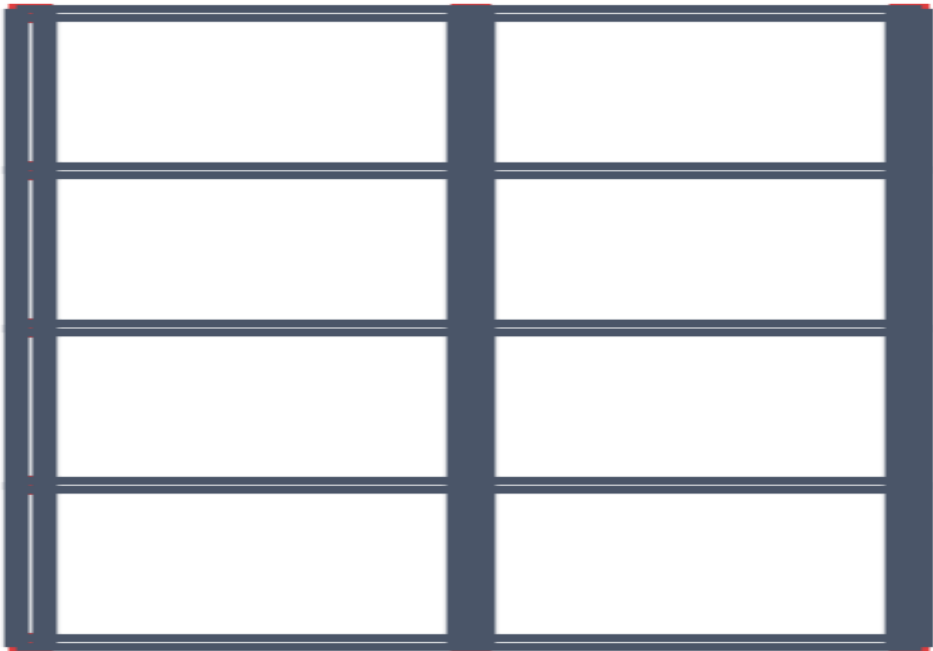
Metric	App Calculated (Real)	Thumb Rule (Expected)	Optimization
Total Steel	4,921 kg (1.27 kg/sqft)	15,500 kg (~4.0 kg/sqft)	68.2% less
Total Concrete	139.1 m3 (0.036 m3/sqft)	155.0 m3 (~0.04 m3/sqft)	10.2% less
Structural Cost	INR 26.1 L	INR 23.2 L (~■600/sqft)	-12.3% cheaper

14 Wide Frontage (Area: 3,875 sqft)



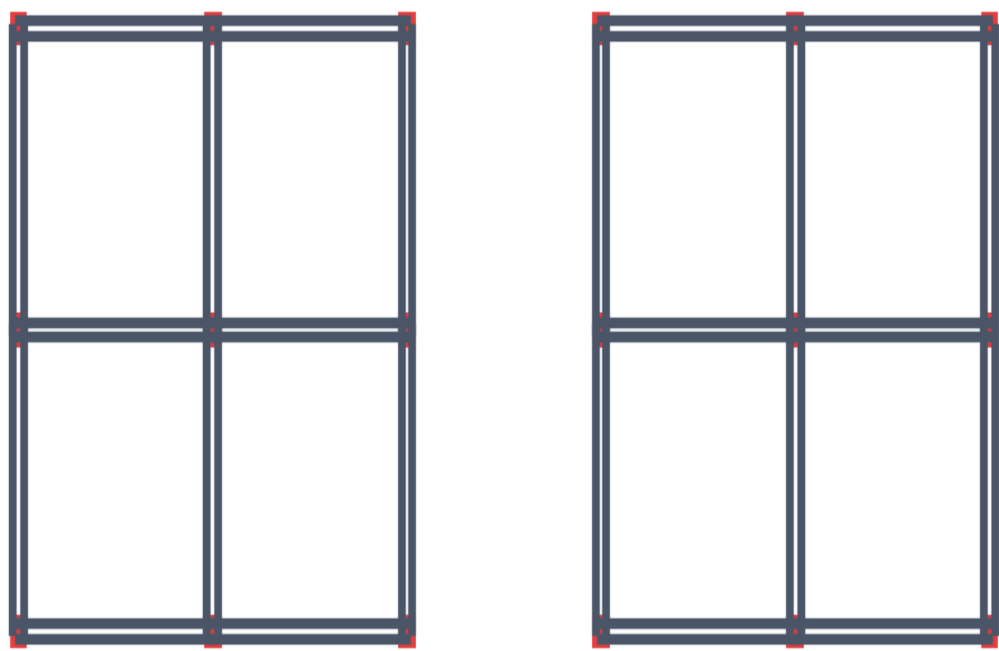
Metric	App Calculated (Real)	Thumb Rule (Expected)	Optimization
Total Steel	5,386 kg (1.39 kg/sqft)	15,500 kg (~4.0 kg/sqft)	65.2% less
Total Concrete	81.4 m3 (0.021 m3/sqft)	155.0 m3 (~0.04 m3/sqft)	47.5% less
Structural Cost	INR 21.4 L	INR 23.2 L (~■600/sqft)	8.0% cheaper

15 Deep Plot (Area: 6,200 sqft)



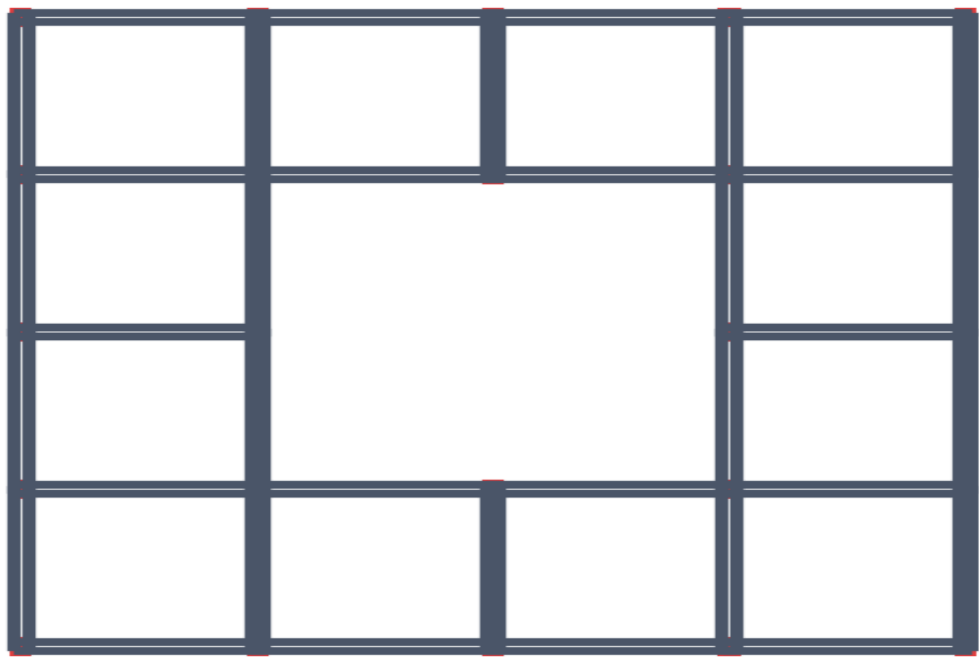
Metric	App Calculated (Real)	Thumb Rule (Expected)	Optimization
Total Steel	6,572 kg (1.06 kg/sqft)	24,800 kg (~4.0 kg/sqft)	73.5% less
Total Concrete	110.4 m3 (0.018 m3/sqft)	248.0 m3 (~0.04 m3/sqft)	55.5% less
Structural Cost	INR 29.6 L	INR 37.2 L (~■600/sqft)	20.4% cheaper

16 Twin Towers (Area: 6,200 sqft)



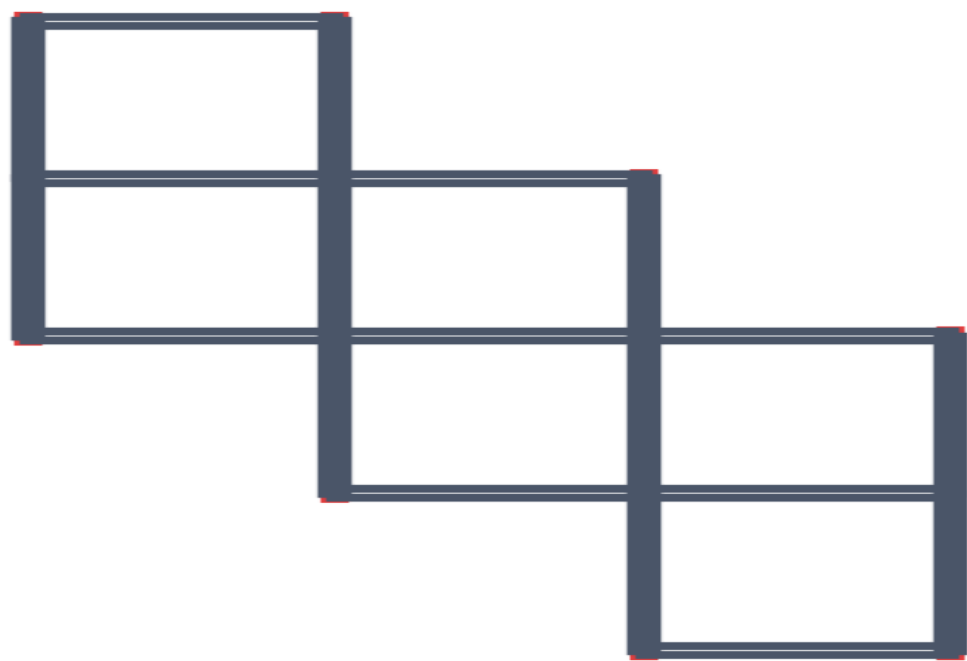
Metric	App Calculated (Real)	Thumb Rule (Expected)	Optimization
Total Steel	6,448 kg (1.04 kg/sqft)	24,800 kg (~4.0 kg/sqft)	74.0% less
Total Concrete	163.1 m3 (0.026 m3/sqft)	248.0 m3 (~0.04 m3/sqft)	34.2% less
Structural Cost	INR 34.6 L	INR 37.2 L (~■600/sqft)	7.0% cheaper

17 O Shape Atrium (Area: 9,300 sqft)



Metric	App Calculated (Real)	Thumb Rule (Expected)	Optimization
Total Steel	9,486 kg (1.02 kg/sqft)	37,200 kg (~4.0 kg/sqft)	74.5% less
Total Concrete	253.9 m3 (0.027 m3/sqft)	372.0 m3 (~0.04 m3/sqft)	31.7% less
Structural Cost	INR 52.6 L	INR 55.8 L (~■600/sqft)	5.7% cheaper

18 Staggered Flats (Area: 4,650 sqft)



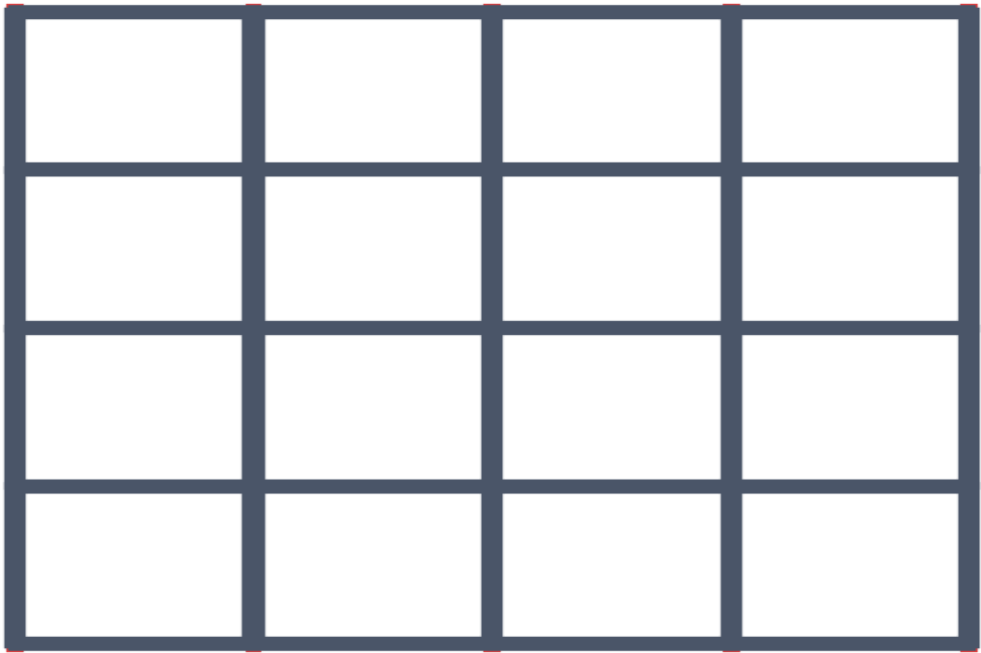
Metric	App Calculated (Real)	Thumb Rule (Expected)	Optimization
Total Steel	6,231 kg (1.34 kg/sqft)	18,600 kg (~4.0 kg/sqft)	66.5% less
Total Concrete	182.7 m3 (0.039 m3/sqft)	186.0 m3 (~0.04 m3/sqft)	1.7% less
Structural Cost	INR 32.8 L	INR 27.9 L (~■600/sqft)	-17.6% cheaper

19 C Shape Clinic (Area: 5,425 sqft)



Metric	App Calculated (Real)	Thumb Rule (Expected)	Optimization
Total Steel	5,805 kg (1.07 kg/sqft)	21,700 kg (~4.0 kg/sqft)	73.2% less
Total Concrete	147.0 m3 (0.027 m3/sqft)	217.0 m3 (~0.04 m3/sqft)	32.3% less
Structural Cost	INR 31.3 L	INR 32.5 L (~■600/sqft)	3.8% cheaper

20 Complex Grid (Area: 20,839 sqft)



Metric	App Calculated (Real)	Thumb Rule (Expected)	Optimization
Total Steel	17,505 kg (0.84 kg/sqft)	83,356 kg (~4.0 kg/sqft)	79.0% less
Total Concrete	358.4 m3 (0.017 m3/sqft)	833.6 m3 (~0.04 m3/sqft)	57.0% less
Structural Cost	INR 79.7 L	INR 125.0 L (~■600/sqft)	36.3% cheaper